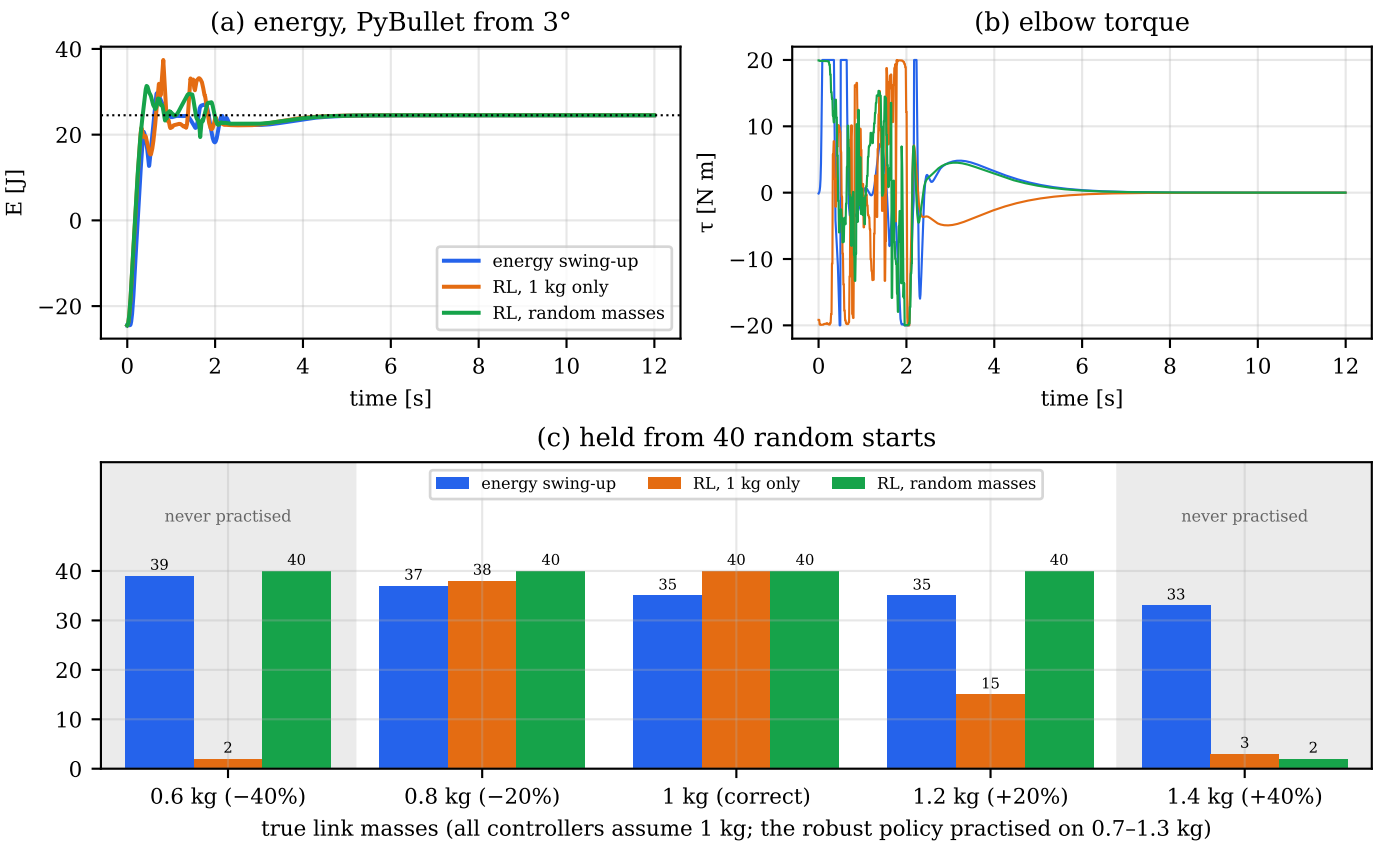
(a) energy, PyBullet from 3°
E [J]
time [s]
energy swing-up
RL, 1 kg only
RL, random masses
(b) elbow torque
τ [N m]
time [s]
(c) held from 40 random starts
never practised
never practised
39
2
40
37
38
40
35
40
40
35
15
40
33
3
2
0.6 kg (−40%)
0.8 kg (−20%)
1 kg (correct)
1.2 kg (+20%)
1.4 kg (+40%)
true link masses (all controllers assume 1 kg; the robust policy practised on 0.7–1.3 kg)